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Readme (Asra)

Hoapa is being designed as a network of small conversations in which one group's work becomes another group's starting point. A question brings participants together; their discussion produces a document that others can examine, challenge, or develop. These documents form a growing map of questions, concepts, arguments, and dependencies. The system gives participants equal standing while allowing their contributions to differ in substance and value. Knowledge acquires structure without that structure becoming a hierarchy of its producers.

The communication protocols specify how these conversations proceed and how their results enter the shared documentation. Software is being built to keep the record and check those rules. Alongside it, a mathematical verification layer is being developed in cubical Agda, a language in which proofs can be checked by a computer. One existing result establishes that consistently renaming participants cannot change the checker's verdict. The broader aim is to make the system's promises precise enough to test and prove.

The mathematical conjecture concerns how such an organization fits the world it investigates. Every inquiry begins somewhere, with particular questions, instruments, assumptions, and things it cannot yet see. Hoapa calls this perspectivism. The challenge is to produce knowledge that other perspectives can use while leaving room for what the original inquiry missed. The project seeks a mathematical correspondence—an adjunction—between what a context makes possible and what an organized practice requires of it. It also asks whether Hoapa's organization can be distinguished from alternatives by its capacity to turn situated work into resources for further inquiry.

The physical starting point is that work depends on constraints that channel energy into particular effects. In living systems, activity also helps maintain the machinery that makes further activity possible. Representations introduce another possibility: an organism, computer, or community can explore consequences in a model before undertaking the corresponding actions in the world. This makes the relation between concrete conditions and abstract operations a powerful resource. Lautman supplies a way to investigate such relations through the mathematical and physical constructions that give them substance; Simondon, Peirce, and Kauffman help connect the question to living beings, their environments, and their practices of understanding. Hoapa's proposed separation between thinkers and the organization of knowledge is an engineering response to this problem.

The project also looks for this pattern inside logic. A proof may begin with a provisional assumption, derive a conclusion, and turn the derivation into a conditional statement that later proofs can use. What began as work under particular conditions becomes a reusable achievement. The harmony between introduction and elimination rules disciplines this passage between establishing something and using it. Hoapa investigates this borrow–work–replay pattern as a possible logical expression of its permacultural ambition.

Autonomous open-science workshops could make these ideas tangible. Participants would investigate questions they care about, compare evidence, construct explanations, and leave documents another workshop could take up. Unresolved disagreements and failed attempts would remain available alongside

successful results. The workshops would also test the organization itself: whether it helps people think together, produces knowledge worth reusing, and sustains their capacity to continue.

Readme (Claude)

The core of my research consists in combining notions from mathematics and its philosophy with ideas from the engineering of networks, at the level of the collective, to develop an extension of the scientific method which incorporates participatory democracy and sustainability. Or more generally, a theory of work constrained by "permacultural" logic: how work can produce reusable results while sustaining the conditions for future work.

The authors it draws on are mathematicians (F.W. Lawvere, A. Grothendieck), philosophers (C.S. Peirce, G. Simondon, G. Deleuze), and scientists (S. Kauffman, J. Bowlby). The main juncture is the philosopher of mathematics Albert Lautman, as read by Fernando Zalamea (who has been supporting this project).

My main thesis states that in a system that leverages the dialectic between concrete and abstract to do some work (brain, computer, community), the relation between concrete substrate and logical labour must be a resource entirely dedicated to the design and application of representations and algorithms. A form of separation between two kinds of decomposition—analytic (as in prime factorization) and synthetic (as in vector coordinates relative to a basis)—is a guiding principle, informed by Lautman, which can keep this relation fruitful: in a community, between thinkers and the knowledge they produce.

The system. Hoapa organizes inquiry as conversations. A question is posed and a small panel is drawn at random. The conversation ends once, either with a result the whole panel accepts or with a recorded failure the next attempt can learn from. Everyone plays by the same rules: the system sorts messages, never people. Results are published anonymously, with their sources attached, so anyone can pick them up, test them or build on them without asking permission. The design is specified, and the software is being built piece by piece.

The verification layer. The protocol has a mathematical twin, checked line by line by a computer (the proof assistant Agda). Already proved:

- each conversation's record reads as a coherent piece of a shared history;
- local agreements assemble into a global one;
- what can be freely copied (knowledge) is kept apart from what gets used up (people's time);
- renaming the participants never changes whether a record is valid, so the checker cannot tell people apart.

The conjecture. The bet is that this way of organizing work is not just one good option but, in a precise mathematical sense, the one best suited to understanding a context and opening it to further inquiry. The candidate tool is an adjunction: a pair of translations, from contexts to the practices they allow and from practices back to what they require, fitted so tightly that every piece of work has an exact counterpart on the other side. Seen this way, there is no view from nowhere, only perspectives and the translations between them, and where they fail to fit together is information, not error.

Work, from physics to Lautman. In physics, work is energy channelled by a constraint, such as a piston or a riverbank. Living things go further: their work rebuilds their own constraints. Some also route it through symbols (genes, neurons, words), where a little work on representations steers a lot of work in the world. That detour is the relation between concrete and abstract, and it should be entirely dedicated to designing

and applying representations and algorithms. Lautman reads mathematics as answers to tensions of this kind (concrete/abstract, synthetic/analytic, structure/existence), and he gives the project its frame. Simondon, Peirce, Bowlby, Grothendieck and Lawvere each add further structure to it.

Permacultural logic. Sustainable work borrows from its context, works, returns something reusable, and leaves the context able to give again. Parts of logic seem built this way. In Gentzen's rules, using a statement never takes out more than proving it put in. A rule that cheats, such as Prior's "tonk", lets anything follow from anything and wrecks the whole system. Making this precise is an invitation to logicians.

Open-science workshops. The first application needs no software. A few volunteers drawn at random get a question, time, and access to experts but no ranks. They publish an answer, or an honest record of their disagreement, that others can reuse. Citizens' assemblies already work this way. Aimed at inquiry, the same format could fit any school or university that values autonomy and learning by doing.